

Telehealth Appointment System

1. Objective

To build a Minimum Viable Product (MVP) web application that allows patients to book, view, and attend virtual video consultations with healthcare providers, enabling basic digital healthcare delivery.

2. Target Users

- **Patients:** Individuals seeking to schedule and attend remote medical consultations.
- **Healthcare Providers (Doctors):** Medical professionals who manage their availability and conduct remote consultations.
- **System Administrators:** Staff responsible for managing user accounts and system configuration.

3. Functional Requirements

- **FR-1: Patient Registration and Authentication**
 - **Description:** The system must allow new patients to create an account using an email address and a password of at least 8 characters.
 - **Validation Conditions:** Email must be unique in the database and follow standard format rules (`user@domain.com`). Password must contain at least one number and one uppercase letter.
 - **Role-Based Conditions:** Accessible only by unauthenticated users.
 - **Example Input:** Email: `patient1@example.com`, Password: `Password123`
 - **Example Output:** Account created successfully, activation email sent to `patient1@example.com`.
 - **Edge Case:** Patient attempts to register with an email that already exists; system returns the error "Email already registered" without creating a duplicate record.
- **FR-2: Provider Availability Management**
 - **Description:** The system must allow authenticated providers to set their weekly recurring availability in 30-minute time slots.
 - **Validation Conditions:** Time slots cannot overlap. Start time must be before end time.
 - **Role-Based Conditions:** Accessible only by authenticated users with the `Provider` role.
 - **Example Input:** Date: `2026-06-01`, Start Time: `09:00 AM`, End Time: `05:00 PM`
 - **Example Output:** Availability saved for `2026-06-01` with 16 available slots.

- **Edge Case:** Provider sets an availability slot in the past; system blocks the action with the error "Cannot set availability in the past".
- **FR-3: Appointment Booking**
 - **Description:** Authenticated patients must be able to select a provider and book an open 30-minute availability slot.
 - **Validation Conditions:** The selected slot must currently be marked as available. The patient cannot have another booking at the same time.
 - **Role-Based Conditions:** Accessible only by authenticated users with the **Patient** role.
 - **Example Input:** Provider ID: **101**, Date: **2026-06-01**, Time Slot: **10:00 AM - 10:30 AM**
 - **Example Output:** Booking confirmed, status updated to **Scheduled**, Slot ID status updated to **Booked**.
 - **Edge Case:** Two patients attempt to book the exact same slot simultaneously; the first request processed succeeds, and the second returns the error "Slot no longer available".
- **FR-4: Appointment Cancellation**
 - **Description:** Patients and providers must be able to cancel scheduled appointments up to 24 hours prior to the appointment start time.
 - **Validation Conditions:** Current time must be more than 24 hours before the appointment start time.
 - **Role-Based Conditions:** Accessible by authenticated users with either **Patient** or **Provider** roles linked to that specific appointment.
 - **Example Input:** Appointment ID: **5001**, User: Patient ID **201**
 - **Example Output:** Appointment status changed to **Cancelled**, time slot status reverted to **Available**.
 - **Edge Case:** User tries to cancel 12 hours before the appointment; system returns error "Cancellations are not allowed within 24 hours of the scheduled time".
- **FR-5: Basic Video Consultation Interface**
 - **Description:** The system must provide a secure video consultation room for scheduled appointments that launches when both participants click "Join".
 - **Validation Conditions:** The "Join" button must only activate 10 minutes prior to the scheduled start time and remain active until the scheduled end time.
 - **Role-Based Conditions:** Only the specific patient and provider assigned to the appointment ID can access the room.
 - **Example Input:** Appointment ID: **5001**, Click event on "Join" button at **09:55 AM** for a **10:00 AM** appointment.
 - **Example Output:** Video room connects successfully, displaying local and remote video streams.
 - **Edge Case:** Patient attempts to join the room 30 minutes before the appointment; the "Join" button is disabled and displays a message "Link activates 10 minutes before the start time".

- **FR-6: Stripe Payment Integration**

- **Description:** The system must require patients to complete payment for the consultation fee using Stripe before the appointment booking is finalized.
- **Validation Conditions:** Payment status must return **Succeeded** from Stripe's API before the appointment status is set to **Scheduled**.
- **Role-Based Conditions:** Patient role during booking workflow.
- **Example Input:** Credit Card Details, Appointment ID: **6001**, Consultation Fee: **\$50.00**.
- **Example Output:** Payment processed successfully, receipt ID **ch_30abc123** generated, appointment status changes to **Scheduled**.
- **Edge Case:** Payment card has insufficient funds; Stripe returns an error code, and the system reverts the time slot to **Available** while notifying the patient.

- **FR-7: SMS and Email Notifications**

- **Description:** The system must automatically send SMS and email reminders to both the provider and the patient 24 hours and 1 hour before the scheduled consultation.
- **Validation Conditions:** Reminders are triggered only for appointments with a status of **Scheduled**.
- **Role-Based Conditions:** System automation service.
- **Example Input:** Appointment ID **7001** scheduled for **2026-06-02 02:00 PM**. Current time is **2026-06-01 02:00 PM**.
- **Example Output:** Email and SMS sent to Patient and Provider; log recorded in the database.
- **Edge Case:** The patient opts out of SMS notifications; the system must skip the SMS trigger and exclusively send the email notification.

- **FR-8: Provider Digital Prescriptions**

- **Description:** During or after the video call, the provider must be able to enter, digitally sign, and issue medical prescriptions to the patient.
- **Validation Conditions:** The prescription must include Patient Name, Medication Name, Dosage, Frequency, and Provider NPI Number. All fields must be non-empty.
- **Role-Based Conditions:** Accessible only by the assigned user with the **Provider** role.
- **Example Input:** Medication: **Amoxicillin**, Dosage: **500mg**, Frequency: **Twice daily for 7 days**.
- **Example Output:** Digital prescription generated, cryptographically signed, and added to the patient's record as a downloadable PDF.
- **Edge Case:** Provider tries to issue a prescription after the appointment status has been marked as **Closed** for more than 48 hours; system blocks action with the message "Prescription window closed".

- **FR-9: Advanced Waiting Room and Connectivity Check**

- **Description:** Patients must complete an automated hardware test (camera, microphone, and internet speed) in a virtual waiting room before joining the consultation.
- **Validation Conditions:** Camera and microphone access must return **Granted** and download speed must be at least **1.5 Mbps**.
- **Role-Based Conditions:** Accessible by the patient up to 15 minutes before the appointment start.
- **Example Input:** System checks audio input device, video input device, and tests a 100KB file download.
- **Example Output:** Hardware status returns **Passed**, and "Enter Consultation Room" button is enabled.
- **Edge Case:** Patient's camera permission is blocked; the system disables the "Enter Consultation Room" button and provides on-screen instructions on how to enable camera access.

4. Non-Functional Requirements

- **NFR-1: Security (Authentication & Authorization)**
 - All user passwords must be hashed using the bcrypt algorithm before storing them in the database.
- **NFR-2: Performance (Page Load Time)**
 - The patient booking dashboard page must load in under 2.0 seconds when tested on a 4G mobile network connection.
- **NFR-3: Reliability (Uptime)**
 - The core appointment booking service must achieve an uptime of 99.5% over any calendar month.
- **NFR-4: Data Compliance (HIPAA)**
 - All stored electronic Protected Health Information (ePHI) must be encrypted at rest using AES-256 encryption.
- **NFR-5: Video Streaming Latency**
 - The peer-to-peer video streaming latency must remain under 300 milliseconds for participants with network connections above 1.5 Mbps.

5. Constraints

- The MVP must run exclusively as a responsive web application supporting the latest versions of Google Chrome, Apple Safari, and Mozilla Firefox.
- The video conferencing capability must be built using open-source WebRTC protocols without custom media server development.